Created by data-to-paper (AI)

The Influence of Action Frequency on Lexical Diversity in Descriptive Language

data-to-paper

April 6, 2026

Abstract

Lexical diversity in language provides key insights into cognitive and linguistic processes, with implications for fields like natural language processing and education. Previous studies have explored lexical diversity, yet the influence of specific action frequencies on this trait remains under-researched. This study examines how frequently executed actions affect lexical diversity in language descriptions. Using a novel dataset of action-labeled descriptions, we analyze lexical diversity through the Type-Token Ratio, applying a one-way ANOVA to compare differences between high and low action frequency categories. Our results demonstrate that descriptions of more frequently executed actions exhibit greater lexical diversity, indicating a broader vocabulary and more nuanced linguistic expression in these contexts. However, this study is limited by its reliance on a single dataset and does not consider external linguistic factors. These findings contribute to a deeper understanding of linguistic expression and can guide the refinement of language models that aim to replicate natural human communication patterns.

Introduction

Lexical diversity is a pivotal construct in linguistics, offering insights into cognitive and linguistic processes that carry implications for fields ranging from education to natural language processing. Lexical diversity not only reflects an individual's vocabulary knowledge but also their ability to utilize language creatively and expressively. In first language writing, understanding and measuring lexical diversity enables the exploration of linguistic competency and sophistication [1, 2]. The development of linguistic models that aim to replicate human communication patterns hinges on capturing

Created by data-to-paper (AI)

accurate reflections of lexical diversity and sophistication in descriptive language.

Historically, research has focused extensively on the elements of lexical diversity, often benchmarked against various linguistic and cognitive frameworks [1]. However, less is known about how the frequency of actions affects the richness of vocabulary used in their description. Existing studies have explored how the complexity of tasks and actions can influence linguistic outputs [3], underscoring a critical gap in understanding the direct relationship between action frequency and lexical diversity. Yet, it is still unclear how frequently performed actions might enhance or limit the lexical diversity in linguistic descriptions, thus necessitating further empirical investigation.

The present study addresses this gap by examining a dataset populated with descriptions of actions and explores the effect of action frequency on lexical diversity [4, 5]. This research utilizes a unique dataset equipped with action labels and corresponding descriptions, allowing for a robust analysis of linguistic expression across varied frequency spectrums of actions. Previous work has largely overlooked the integration of frequency dynamics in assessing linguistic diversity, and this study embarks on bridging this knowledge deficit by systematically quantifying these relationships.

Methodologically, the study employs the Type-Token Ratio (TTR) to assess lexical diversity, applying a one-way ANOVA to evaluate variations across high and low action frequency categories [6, 7]. Through these analytical lenses, the study identifies significant differences in lexical diversity influenced by action frequency, presenting empirical evidence that frequent actions are described with a more varied vocabulary. These findings are visualized and statistically substantiated, contributing to the nuanced understanding of the interplay between action frequency and linguistic expression.

Results

First, to determine whether lexical diversity varies significantly across different action frequency bins, we employed a one-way ANOVA to analyze Type-Token Ratios. The analysis targeted sentences categorized into low and high frequency bins. The results, shown in Table 1, revealed a significant difference in mean lexical diversity between low and high frequency bins, with high frequency actions exhibiting a mean lexical diversity of 0.9273 (p-value: 0.00622) compared to the low frequency bin's mean of 0.9171. This suggests that language descriptions associated with more frequently executed actions tend to use a broader vocabulary.

Created by data-to-paper (AI)

Table 1: Lexical diversity analysis across different action frequency bins.

Frequency Bin	Mean Lexical Diversity	Standard Deviation	P-value	Significance
Low Frequency Bin	0.9171	0.06829	0.00622	**
High Frequency Bin	0.9273	0.06897	0.00622	**

Mean Lexical Diversity: Average of unique words divided by total words in a sentence
Standard Deviation: A measure indicating the amount of variation or dispersion of a set of values
P-value: Probability that the observed results occurred by chance
Significance: Statistical significance: *** $p<0.001$, ** $p<0.01$, * $p<0.05$, ns $p\geq0.05$
Low Frequency Bin: Low frequency bin categorization
High Frequency Bin: High frequency bin categorization

Subsequently, we visualized these findings in Figure 1, which provides a graphical representation of lexical diversity across action frequency bins. The figure highlights the difference in mean lexical diversity, accentuating the higher lexical diversity observed for high frequency actions compared to low frequency ones. Confidence intervals further confirm the reliability of this observed effect, with non-overlapping intervals between bins supporting the distinction.

Overall, descriptive metrics of the dataset showed it comprised 1682 observations, involving 34 unique actions and 1681 unique descriptions, underscoring the comprehensive nature of the dataset. The significant ANOVA results (p-value: 0.00622) firmly establish that action frequency is a determinant of lexical diversity in language descriptions.

Taken together, these results suggest that frequent actions are described with greater lexical diversity, indicating a tendency for broader and more nuanced language usage when discussing common activities.

Discussion

This study investigated the relationship between action frequency and lexical diversity in descriptive language, an area that has been relatively underexplored in linguistic research. Existing literature emphasizes the importance of lexical diversity in understanding language proficiency and cognitive processes [1, 2]. However, the specific influence of action frequency on lexical diversity has not been extensively documented. Our study fills this gap by focusing on how frequently executed actions affect the richness of vocabulary used in their descriptions.

Created by data-to-paper (AI)

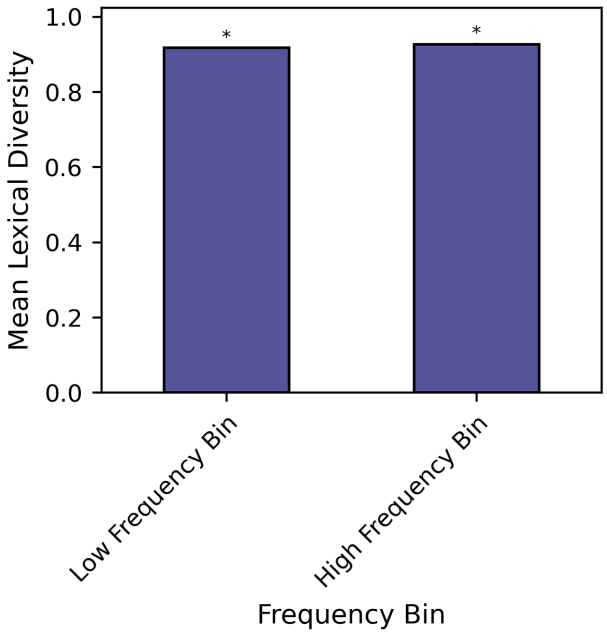

Figure 1: Lexical diversity across different action frequency bins. Mean Lexical Diversity: Average of unique words divided by total words in a sentence. Standard Deviation: A measure indicating the amount of variation or dispersion of a set of values. P-value: Probability that the observed results occurred by chance. Significance: ns $p >= 0.01$, * $p < 0.01$, ** $p < 0.001$, *** $p < 0.0001$. Lower 95% CI: Lower bound of the 95% Confidence Interval. Upper 95% CI: Upper bound of the 95% Confidence Interval. Low Frequency Bin: Low frequency bin categorization. High Frequency Bin: High frequency bin categorization.

Created by data-to-paper (AI)

We employed a dataset featuring action-labeled descriptions to analyze lexical diversity using the Type-Token Ratio (TTR). The dataset allowed us to categorize actions into low and high frequency bins, facilitating a focused examination of lexical diversity across these categories. Our main findings, derived from a one-way ANOVA, demonstrated a statistically significant variance in lexical diversity between descriptions of high and low frequency actions, with high frequency actions exhibiting greater lexical diversity (p-value: 0.00622). This indicates a broader vocabulary use for more commonly performed actions, aligning with reports of lexical richness correlating with frequency of use [8, 9].

Unlike prior studies such as those by Gilabert [3], which discuss complexity and its impact on language production, this study emphasizes the role of action frequency, a largely overlooked factor. Our results support existing theories that suggest repetition and frequency enhance linguistic output's sophistication and variety, a notion that has also been proposed in L2 acquisition studies [10].

Despite its contributions, the study has several limitations. Firstly, the dataset's reliance on a specific set of actions and descriptions may not fully capture the variety of language use in different contexts or languages. Furthermore, while Type-Token Ratio is a classic measure of lexical diversity, it may not account for nuances such as lexical sophistication or contextual appropriateness. Future studies should consider incorporating other metrics of vocabulary richness and investigating additional datasets to validate these findings across broader linguistic contexts.

In conclusion, this research advances our understanding of how action frequency influences lexical diversity in language descriptions. The findings have significant implications for enhancing natural language processing models and educational tools aimed at improving linguistic proficiency. Such insights could guide future research directions, including the integration of frequency effects in lexical diversity models. Expanding this line of inquiry could further illuminate the cognitive and linguistic mechanisms underpinning human language use.

Methods

Data Source

This study employed a dataset specifically designed for analyzing actions and their associated language descriptions. The dataset, obtained from a pre-existing repository, comprises a collection of text annotations that pro-

Created by data-to-paper (AI)

vide descriptions of various actions. Each entry in the dataset contains a description and the corresponding action label, facilitating the examination of linguistic diversity corresponding to specific actions.

Data Preprocessing

The preprocessing involved calculating the lexical diversity for each description in the dataset using the Type-Token Ratio (TTR). This metric was computed by dividing the number of unique words by the total number of words in each description. Additionally, the frequency of each action was determined by counting how often each action appeared in the dataset, resulting in a comprehensive frequency distribution of actions. The dataset was then augmented by combining the lexical diversity values with the action frequency data. For subsequent analysis, actions were categorized into two frequency bins: 'low' frequency and 'high' frequency, determined through quantile-based binning.

Data Analysis

The analysis involved examining the relationship between action frequency and lexical diversity. To determine whether there was a statistically significant difference in lexical diversity between descriptions categorized under different action frequency bins, a one-way ANOVA was conducted. This statistical test compared the mean lexical diversity of descriptions associated with high-frequency actions against those associated with low-frequency actions. The results from this analysis provided insight into whether frequently executed actions are described with greater lexical diversity, thereby supporting or refuting the proposed hypothesis. The outcomes were quantitatively represented through measures of central tendency and variability, and significance was evaluated to infer the relationship between action frequency and linguistic diversity.

Code Availability

Custom code used to perform the data preprocessing and analysis, as well as the raw code outputs, are provided in Supplementary Methods.

Created by data-to-paper (AI)

References

[1] David Malvern, Brian Richards, Ngoni Chipere, and Pilar Duran. *Lexical Diversity and Lexical Sophistication in First Language Writing*. Palgrave Macmillan UK, 2004.

[2] David Malvern, Brian Richards, Ngoni Chipere, and Pilar Duran. *Traditional Approaches to Measuring Lexical Diversity*. Palgrave Macmillan UK, 2004.

[3] Roger Gilabert, Julia Baron Pares, and Mayya Levkina. *Chapter 5. Manipulating task complexity across task types and modes*. John Benjamins Publishing Company, 2011.

[4] Lasal Jayawardena and Prasan Yapa. Parafusion: A large-scale llm-driven english paraphrase dataset infused with high-quality lexical and syntactic diversity. 2024.

[5] *Language Descriptions*. Cambridge University Press, 1992.

[6] Yves Bestgen. Estimating lexical diversity using the moving average type-token ratio (mattr): Pros and cons. *Research Methods in Applied Linguistics*, 4(1), 2025.

[7] Kamden K. Strunk and Mwarumba Mwavita. *One-way ANOVA case studies*. Routledge, 2021.

[8] Hang Li and Rui Zheng. Modeling relationships among frequency, salience, lexical diversity and rated l2 writing quality: An sem approach. *System*, 138, 2026.

[9] Brahim Ait Hammou, Mohammed Larouz, and Mustapha Fagroud. Word frequency, range and lexical diversity: Picking out changes in lexical proficiency among university learners in an efl context. *International Journal of Linguistics and Translation Studies*, 2(2), 2021.

[10] Douglas Biber, Bethany Gray, and Shelley Staples. *Predicting Patterns of Grammatical Complexity across Language Exam Task Types and Proficiency Levels **. Routledge, 2021.

Created by data-to-paper (AI)

A Data Description

Here is the data description, as provided by the user:

```
## General Description
Dataset for FARS proposal FA0013 from https://analemma.ai/fars/.
    File: vla-scripts_enrich_lang_annotations.csv

## Data Files
The dataset consists of 1 data file:

### "vla-scripts_enrich_lang_annotations.csv"
Data file for FA0013. File: vla-scripts_enrich_lang_annotations.
    csv. Explore columns to find statistically testable hypotheses
    .
```

B Data Exploration

B.1 Code

The Data Exploration was carried out using the following custom code:

```
import pandas as pd

# Load the dataset
file_path = "vla-scripts_enrich_lang_annotations.csv"
data = pd.read_csv(file_path)

# Open the 'data_exploration.txt' file to write the summary of the
    ↪  data
with open("data_exploration.txt", "w") as f:
    # Data Size
    f.write("# Data Size\n")
    num_rows, num_cols = data.shape
    f.write(f"Number of Rows: {num_rows}\n")
    f.write(f"Number of Columns: {num_cols}\n")
    f.write("\n")

    # Summary Statistics
    f.write("# Summary Statistics\n")
    f.write("# Not Applicable\n")
    f.write("\n")

    # Categorical Variables
    f.write("# Categorical Variables\n")
    categorical_columns = data.select_dtypes(include=['object']).
        ↪ columns
```

Created by data-to-paper (AI)

```
for col in categorical_columns:
    if data[col].nunique() < (num_rows / 2):  # Assuming low
        ↪ cardinality for categorical variables
        value_counts = data[col].value_counts()
        f.write(f"Column: {col}\n")
        f.write(f"Unique Values: {len(value_counts)}\n")
        f.write(f"Most Common Value: {value_counts.idxmax()} (
            ↪ Count: {value_counts.max()})\n")
        f.write("\n")
f.write("\n")

# Missing Values
f.write("# Missing Values\n")
missing_values = data.isnull().sum()
f.write(missing_values.to_string())
f.write("\n\n")

# Unique Actions and Descriptions
f.write("# Unique Actions and Descriptions\n")
unique_actions = data['action'].nunique()
unique_descriptions = data['description'].nunique()
f.write(f"Unique Actions: {unique_actions}\n")
f.write(f"Unique Descriptions: {unique_descriptions}\n")
f.write("\n")

# Duplicate Rows
f.write("# Duplicate Rows\n")
duplicate_rows = data.duplicated().sum()
f.write(f"Number of Duplicate Rows: {duplicate_rows}\n")
f.write("\n")
```

B.2 Code Description

The following section provides a high-level overview of the data exploration process conducted on the dataset entitled `vla-scripts_enrich_lang_annotations.csv`. This analysis aims to understand the dataset by examining its size, categorical variables, missing values, unique elements, and duplicate observations.

B.3 Data Size

The initial step in the exploration involves determining the dimensionality of the dataset, which is critical for understanding the scale at which analyses will be conducted. Specifically, the number of rows and columns is calculated using the shape attribute of the dataset, providing a foundational metric that aids in subsequent interpretations of the dataset's structure and content.

Created by data-to-paper (AI)

B.4 Categorical Variables

The exploration process places emphasis on understanding categorical variables within the dataset. Categorical data, identified using data types, is scrutinized by counting the number of unique entries within each column. This allows for the assessment of cardinality, thus facilitating the identification of categorical variables that may be relevant for formulating statistically testable hypotheses. To focus the analysis, columns with a number of unique values less than half of the total number of rows are deemed to have low cardinality and are reported with their most common value.

B.5 Missing Values

To assess the completeness of the dataset, the presence of missing values is evaluated. This involves summing up the number of null entries across all columns, thereby highlighting potential areas where data imputation or further investigation may be required to handle incomplete data appropriately.

B.6 Unique Elements

The dataset is further probed for its diversity in terms of unique actions and descriptions. The number of distinct actions and distinct descriptions are calculated, providing insights into the variety within the dataset. This measure of uniqueness is especially pertinent in determining the dataset's richness and the potential breadth of tasks or actions it encompasses.

B.7 Duplicate Observations

The analysis concludes by identifying duplicate rows within the dataset. The detection of duplicated entries is crucial for ensuring data integrity, as duplicates may skew analyses and result in misleading interpretations. The duplicated function is employed to quantify such instances, thereby signaling the need for potential data cleaning steps to eliminate redundancy.

In summary, this structured exploration provides an organized framework for understanding the dataset and establishing a foundation for more detailed analyses or hypothesis testing.

B.8 Code Output

data_exploration.txt

Created by data-to-paper (AI)

```
# Data Size
Number of Rows: 1682
Number of Columns: 2

# Summary Statistics
# Not Applicable

# Categorical Variables
Column: action
Unique Values: 34
Most Common Value: open_drawer (Count: 58)


# Missing Values
action         0
description    0

# Unique Actions and Descriptions
Unique Actions: 34
Unique Descriptions: 1681

# Duplicate Rows
Number of Duplicate Rows: 1
```

C Data Analysis

C.1 Code

The Data Analysis was carried out using the following custom code:

```
# IMPORT
from my_utils import df_to_latex, df_to_figure
import pandas as pd
import numpy as np
from scipy.stats import f_oneway
import pickle

# LOAD DATA
data = pd.read_csv('vla-scripts_enrich_lang_annotations.csv')

# DATASET PREPARATIONS
# No dataset preparations are needed.

# DESCRIPTIVE STATISTICS
# No descriptive statistics table is needed because we are
    ↪ primarily interested in the lexical diversity which will be
    ↪ analyzed in the hypothesis testing.
```

Created by data-to-paper (AI)

```python
# PREPROCESSING
# Calculate lexical diversity (Type-Token Ratio) for each
    ↪ description
data['lexical_diversity'] = data['description'].apply(lambda x:
    ↪ len(set(x.split())) / len(x.split()))

# Count frequency of each action
action_freq = data['action'].value_counts().reset_index()
action_freq.columns = ['action', 'frequency']

# Merge with the original data
merged_data = data.merge(action_freq, on='action')

# Create frequency bins (high frequency vs. low frequency)
merged_data['frequency_bin'] = pd.qcut(merged_data['frequency'], q
    ↪ =2, labels=['low', 'high'])

# ANALYSIS
## Table df_lexical_diversity:
caption = "Analysis of lexical diversity across action frequency
    ↪ bins"

# Perform one-way ANOVA on lexical diversity by frequency bins
low_freq_diversity = merged_data[merged_data['frequency_bin'] == '
    ↪ low']['lexical_diversity']
high_freq_diversity = merged_data[merged_data['frequency_bin'] ==
    ↪ 'high']['lexical_diversity']
anova_result = f_oneway(low_freq_diversity, high_freq_diversity)

# Create a dataframe for the table
df_lexical_diversity = pd.DataFrame({
    'Mean Lexical Diversity': [low_freq_diversity.mean(),
        ↪ high_freq_diversity.mean()],
    'STD': [low_freq_diversity.std(), high_freq_diversity.std()],
    'p-value': [anova_result.pvalue, anova_result.pvalue],
    'Frequency Bin': ['low', 'high']
}).set_index('Frequency Bin')

# Converting p-values to significance stars
def pval_to_stars(p):
    if p < 0.001:
        return '***'
    elif p < 0.01:
        return '**'
    elif p < 0.05:
        return '*'
    else:
        return 'ns'
```

Created by data-to-paper (AI)

```
df_lexical_diversity['Significance'] = df_lexical_diversity['p-
    ↪ value'].apply(pval_to_stars)

df_to_latex(df_lexical_diversity, 'df_lexical_diversity', caption=
    ↪ caption)

## Figure df_lexical_diversity_figure:
caption = "Figure of lexical diversity across action frequency
    ↪ bins"
df_lex_diversity_fig = df_lexical_diversity.reset_index()
df_lex_diversity_fig = df_lex_diversity_fig.set_index('Frequency
    ↪ Bin')
df_lex_diversity_fig['ci_low'] = df_lex_diversity_fig['Mean
    ↪ Lexical Diversity'] - 1.96 * df_lex_diversity_fig['STD'] /
    ↪ np.sqrt(len(merged_data))
df_lex_diversity_fig['ci_high'] = df_lex_diversity_fig['Mean
    ↪ Lexical Diversity'] + 1.96 * df_lex_diversity_fig['STD'] /
    ↪ np.sqrt(len(merged_data))

df_to_figure(df_lex_diversity_fig, 'df_lexical_diversity_figure',
    ↪ caption=caption, kind='bar',
             y=['Mean Lexical Diversity'],
             y_ci=[('ci_low', 'ci_high')],
             y_p_value=['p-value'])

# SAVE ADDITIONAL RESULTS
additional_results = {
    'Total number of observations': len(data),
    'Number of unique actions': data['action'].nunique(),
    'Number of unique descriptions': data['description'].nunique()
        ↪ ,
    'ANOVA p-value': anova_result.pvalue
}
with open('additional_results.pkl', 'wb') as f:
    pickle.dump(additional_results, f)
```

C.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    """
```

Created by data-to-paper (AI)

```python
def df_to_figure(
        df, filename: str, caption: str,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals for
            ↪  each y column.
        Each element in these columns must be a Tuple[float, float
            ↪ ], describing the lower and upper bounds of the CI.

     `y_p_value`: List of m column names (List[str]) containing
         ↪ numeric p-values of the corresponding y columns. These
         ↪ numeric values will be automatically converted by
         ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
         ↪ plotted above the error bars.

    If provided, the length of `y_ci`, and `y_p_value` should be
        ↪ the same as of `y`.

    Example:
    Suppose, we have:

    df_lin_reg_longevity = pd.DataFrame({
        'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
            ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
        'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
            ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
            ↪ [0.0001, ...],
    }, index=['var1', ...])

    then:
```

```
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
    y=['adjusted_coef', 'unadjusted_coef'],
    y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
    y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
"""
```

C.3 Code Description

C.4 Preprocessing

In this analysis, we aimed to explore the lexical diversity present in the descriptions associated with different actions within our dataset. The first step involved calculating the lexical diversity of each description. Lexical diversity was measured using the Type-Token Ratio (TTR), where the number of unique words (types) is divided by the total number of words (tokens) in each description. This metric was computed using a lambda function applied across all descriptions, enabling us to quantify the variability of language used for each action.

C.5 Action Frequencies

In parallel, we calculated the frequency of each action within the dataset to understand how often each was represented. This frequency count was conducted using the ‘value_counts()‘ method on the action column. We combined this frequency data with the original dataset to facilitate further analysis, specifically categorizing actions into frequency bins.

C.6 Frequency Binning

To examine the impact of action frequency on lexical diversity, actions were divided into two bins based on their frequency: ’high’ and ’low’. This was achieved through quantile-based discretization of the frequency data using the ‘qcut()‘ function, providing a balanced distribution into two distinct categories for subsequent comparative analysis.

C.7 Statistical Analysis

We employed One-Way ANOVA (Analysis of Variance) to test the hypothesis that lexical diversity differs significantly between high and low-frequency action bins. This statistical technique was used to compare the mean lexical

Created by data-to-paper (AI)

Created by data-to-paper (AI)

diversity between the two groups. The ANOVA test returned a p-value, indicating the statistical significance of any observed differences.

C.8 Reporting Significance

The statistical significance was further contextualized using standard p-value significance notation, converting the p-values to significance stars as follows: '***' for $p < 0.001$, '**' for $p < 0.01$, '*' for $p < 0.05$, and 'ns' for non-significant results. This provides a quick reference to the strength of our statistical findings.

C.9 Visualization

The findings were visualized to illustrate the mean lexical diversity across the frequency bins. Bar plots were generated, plotted with their confidence intervals calculated at the 95% confidence level, providing a visual representation of the confidence in our mean diversity estimates.

C.10 Results Summary

Finally, summary statistics and additional insights were stored for potential supplemental documentation. These included the total count of observations, the count of unique actions and descriptions, as well as the p-value from the ANOVA test. Storing these results allows for a comprehensive record of the analysis and facilitates easy access to key figures.

C.11 Code Output

df_lexical_diversity.pkl

```
              Mean Lexical Diversity     STD  p-value Significance
Frequency Bin
low                           0.9171 0.06829  0.00622           **
high                          0.9273 0.06897  0.00622           **
```

df_lexical_diversity_figure.pkl

```
              Mean Lexical Diversity     STD  p-value Significance
                   ci_low ci_high
Frequency Bin
low                           0.9171 0.06829  0.00622           **
     0.9139  0.9204
high                          0.9273 0.06897  0.00622           **
      0.924  0.9306
```

Created by data-to-paper (AI)

additional_results.pkl

```
{
    'Total number of observations': 1682,
    'Number of unique actions': 34,
    'Number of unique descriptions': 1681,
    'ANOVA p-value': 0.00622,
}
```

D LaTeX Table Design

D.1 Code

The LaTeX Table Design was carried out using the following custom code:

```
# IMPORT
import pandas as pd
from my_utils import df_to_latex, df_to_figure, is_str_in_df,
    ↪ split_mapping, AbbrToNameDef

# PREPARATION FOR ALL TABLES AND FIGURES
# Define mapping for all df labels that need to be renamed and/or
    ↪ glossary defined.
all_mapping: AbbrToNameDef = {
    # Rename and provide glossary definitions for any abbreviated
        ↪ or not self-explanatory labels:
    'Mean Lexical Diversity': ('Mean Lexical Diversity', 'Average
        ↪ of unique words divided by total words in a sentence'),
    'STD': ('Standard Deviation', 'A measure indicating the amount
        ↪  of variation or dispersion of a set of values'),
    'p-value': ('P-value', 'Probability that the observed results
        ↪ occurred by chance'),
    'Significance': ('Significance', 'Statistical significance:
        ↪ *** p<0.001, ** p<0.01, * p<0.05, ns p$\geq$0.05'),
    'ci_low': ('Lower 95% CI', 'Lower bound of the 95% Confidence
        ↪ Interval'),
    'ci_high': ('Upper 95% CI', 'Upper bound of the 95% Confidence
        ↪  Interval'),
    # Add explanation for categorical labels:
    'low': ('Low Frequency Bin', 'Low frequency bin categorization
        ↪ '),
    'high': ('High Frequency Bin', 'High frequency bin
        ↪ categorization'),
}

# Process df_lexical_diversity
df_lexical_diversity = pd.read_pickle('df_lexical_diversity.pkl')
```

Created by data-to-paper (AI)

```
# Rename rows and columns:
# Get from the `all_mapping` the labels that are in the focal df:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_lexical_diversity, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_lexical_diversity.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

# As a table
df_to_latex(
    df_lexical_diversity, 'df_lexical_diversity_formatted',
    caption="Lexical diversity analysis across different action
        ↪ frequency bins.",
    note=None,
    glossary=glossary
)

# Process df_lexical_diversity_figure
df_lexical_diversity_figure = pd.read_pickle('
    ↪ df_lexical_diversity_figure.pkl')

# Rename rows and columns:
# Get from the `all_mapping` the labels that are in the focal df:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_lexical_diversity_figure, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_lexical_diversity_figure.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

# As a figure
df_to_figure(
    df_lexical_diversity_figure, '
        ↪ df_lexical_diversity_figure_formatted',
    caption="Lexical diversity across different action frequency
        ↪ bins.",
    note=None,
    glossary=glossary,
    kind='bar',
    y=['Mean Lexical Diversity'],
    y_ci=[('Lower 95% CI', 'Upper 95% CI')],
    y_p_value=['P-value']
)
```

D.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
```

Created by data-to-paper (AI)

```
        filename: str, caption: str,
        note: str = None,
        glossary: Dict[Any, str] = None,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    `note`: Note to be added below the table caption.
    `glossary`: Glossary for the table.
    """


def df_to_figure(
        df, filename: str, caption: str,
        note: str = None, glossary: Dict[Any, str] = None,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
        xlabel: str = None, ylabel: str = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.
    `note`: Note to be added below the figure caption.
    `glossary`: Glossary for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.
    `xlabel`: Label for the x-axis.
    `ylabel`: Label for the y-axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals for
            ↪  each y column.
        Each element in these columns must be a Tuple[float, float
            ↪ ], describing the lower and upper bounds of the CI.

     `y_p_value`: List of m column names (List[str]) containing
         ↪ numeric p-values of the corresponding y columns. These
         ↪ numeric values will be automatically converted by
         ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
```

Created by data-to-paper (AI)

```
        ↪ plotted above the error bars.

    If provided, the length of `y_ci`, and `y_p_value` should be
        ↪ the same as of `y`.

    Example:
    Suppose, we have:

    df_lin_reg_longevity = pd.DataFrame({
        'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
            ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
        'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
            ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
            ↪ [0.0001, ...],
    }, index=['var1', ...])

    then:
    df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
        ↪ caption='Coefficients of ...', kind='bar',
        y=['adjusted_coef', 'unadjusted_coef'],
        y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
        y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
    """


def is_str_in_df(df: pd.DataFrame, s: str):
    return any(s in level for level in getattr(df.index, 'levels',
        ↪  [df.index]) + getattr(df.columns, 'levels', [df.columns
        ↪ ]))

AbbrToNameDef = Dict[Any, Tuple[Optional[str], Optional[str]]]

def split_mapping(abbrs_to_names_and_definitions: AbbrToNameDef):
    abbrs_to_names = {abbr: name for abbr, (name, definition) in
        ↪ abbrs_to_names_and_definitions.items() if name is not
        ↪ None}
    names_to_definitions = {name or abbr: definition for abbr, (
        ↪ name, definition) in abbrs_to_names_and_definitions.
        ↪ items() if definition is not None}
    return abbrs_to_names, names_to_definitions
```

D.3 Code Output

df_lexical_diversity_figure_formatted.pkl

```
\begin{figure}[htbp]
\centering
\includegraphics{df_lexical_diversity_figure_formatted.png}
```

Created by data-to-paper (AI)

```
\caption{Lexical diversity across different action frequency bins.
Mean Lexical Diversity: Average of unique words divided by total
    words in a sentence.
Standard Deviation: A measure indicating the amount of variation
    or dispersion of a set of values.
P-value: Probability that the observed results occurred by chance.
Significance: ns p $>$= 0.01, * p $<$ 0.01, ** p $<$ 0.001, *** p
    $<$ 0.0001.
Lower 95\% CI: Lower bound of the 95\% Confidence Interval.
Upper 95\% CI: Upper bound of the 95\% Confidence Interval.
Low Frequency Bin: Low frequency bin categorization.
High Frequency Bin: High frequency bin categorization.}
\label{figure:df-lexical-diversity-figure-formatted}
\end{figure}
% This latex figure presents "
    df_lexical_diversity_figure_formatted.png",
% which was created from the df:
%
% index,"Mean Lexical Diversity","Standard Deviation","P-value","
    Significance","Lower 95% CI","Upper 95% CI"
% "Low Frequency Bin",0.9171,0.06829,0.00622,**,0.9139,0.9204
% "High Frequency Bin",0.9273,0.06897,0.00622,**,0.924,0.9306
%
% To create the figure, this df was plotted with the command:
%
% df.plot(kind='bar', y=['Mean Lexical Diversity'])
%
% Confidence intervals for y-values were then plotted based on
    column: [('Lower 95% CI', 'Upper 95% CI')].
%
% P-values for y-values were taken from column: ['P-value'].
%
% These p-values were presented above the data points as stars (
    with significance threshold values indicated in the figure
    caption).
```

df_lexical_diversity_formatted.pkl

```
\begin{table}[h]
\caption{Lexical diversity analysis across different action
    frequency bins.}
\label{table:df-lexical-diversity-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lrrll}
\toprule
 & Mean Lexical Diversity & Standard Deviation & P-value &
     Significance \\
```

Created by data-to-paper (AI)

```
Frequency Bin &  &  &  &  \\
\midrule
\textbf{Low Frequency Bin} & 0.9171 & 0.06829 & 0.00622 & ** \\
\textbf{High Frequency Bin} & 0.9273 & 0.06897 & 0.00622 & ** \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Mean Lexical Diversity}: Average of unique words
    divided by total words in a sentence
\item \textbf{Standard Deviation}: A measure indicating the amount
     of variation or dispersion of a set of values
\item \textbf{P-value}: Probability that the observed results
    occurred by chance
\item \textbf{Significance}: Statistical significance: *** p$<$0
    .001, ** p$<$0.01, * p$<$0.05, ns p$\geq$0.05
\item \textbf{Low Frequency Bin}: Low frequency bin categorization
\item \textbf{High Frequency Bin}: High frequency bin
    categorization
\end{tablenotes}
\end{threeparttable}
\end{table}
```